# The correlation between Driving Hours and Trucks Fatal Accident

Thanh Nguyen - DS220 - Fall 2023

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# Introduction

To support the "Hours of Service" regulations, the University of Michigan Transportation Research Institute manages a dataset containing information about all medium- and heavy-duty trucks that have been involved in fatal accidents in the US

"The Trucks in Fatal Accidents database (TIFA) contain records for all the medium and heavy trucks that were involved in fatal traffic crashes in the 50 states and District of Columbia"

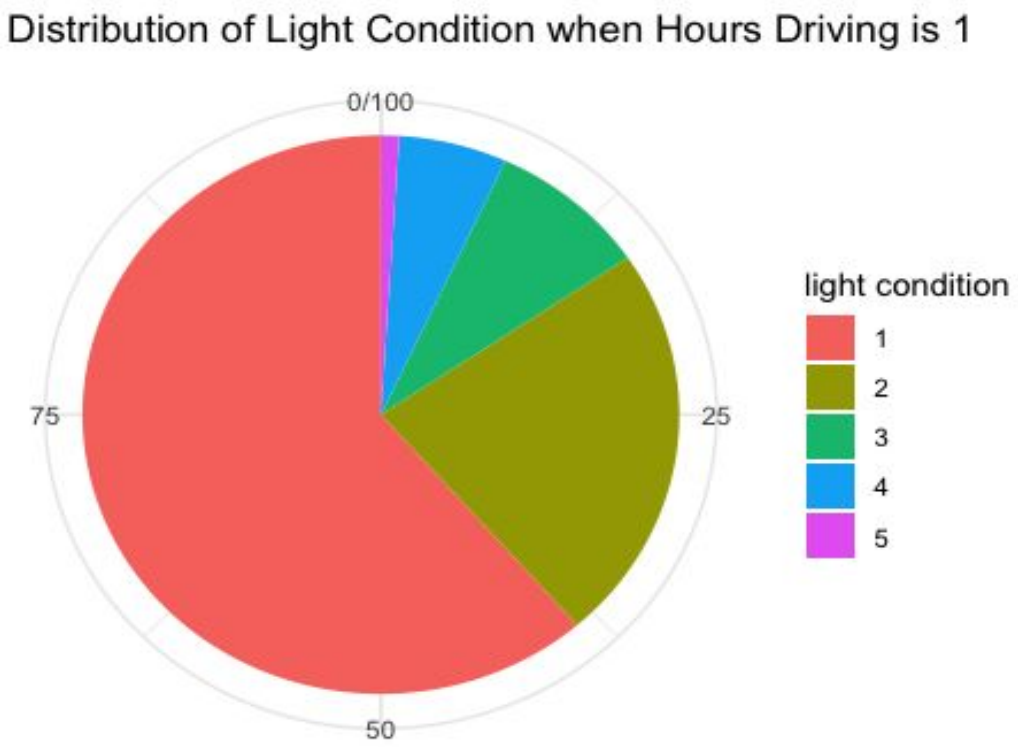
- TIFA 1999 - 2007
- Overview

# Research Question

Does the number of continuous hours that a truck driver can operate his or her vehicle without sleep impact the number of fatal accidents?

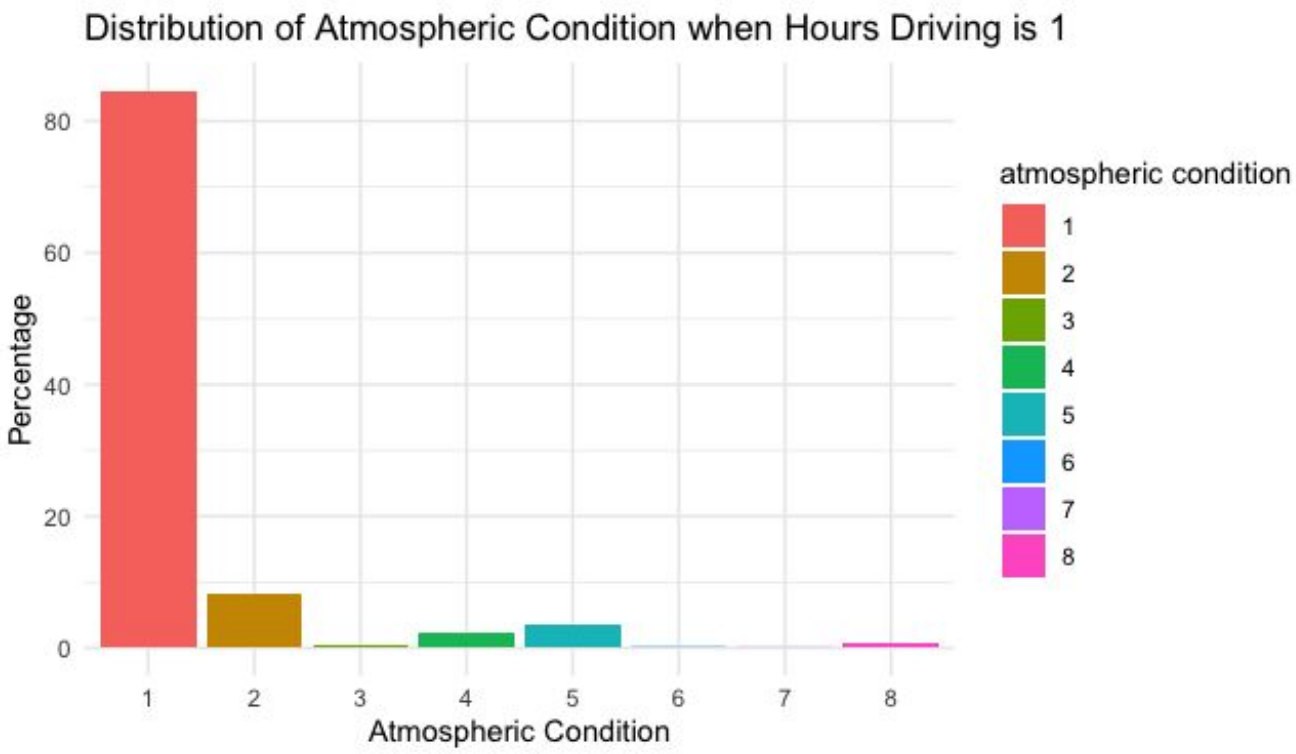
# Accident Cases in Continuous Driving Hours

# Light Condition when Accidents happened



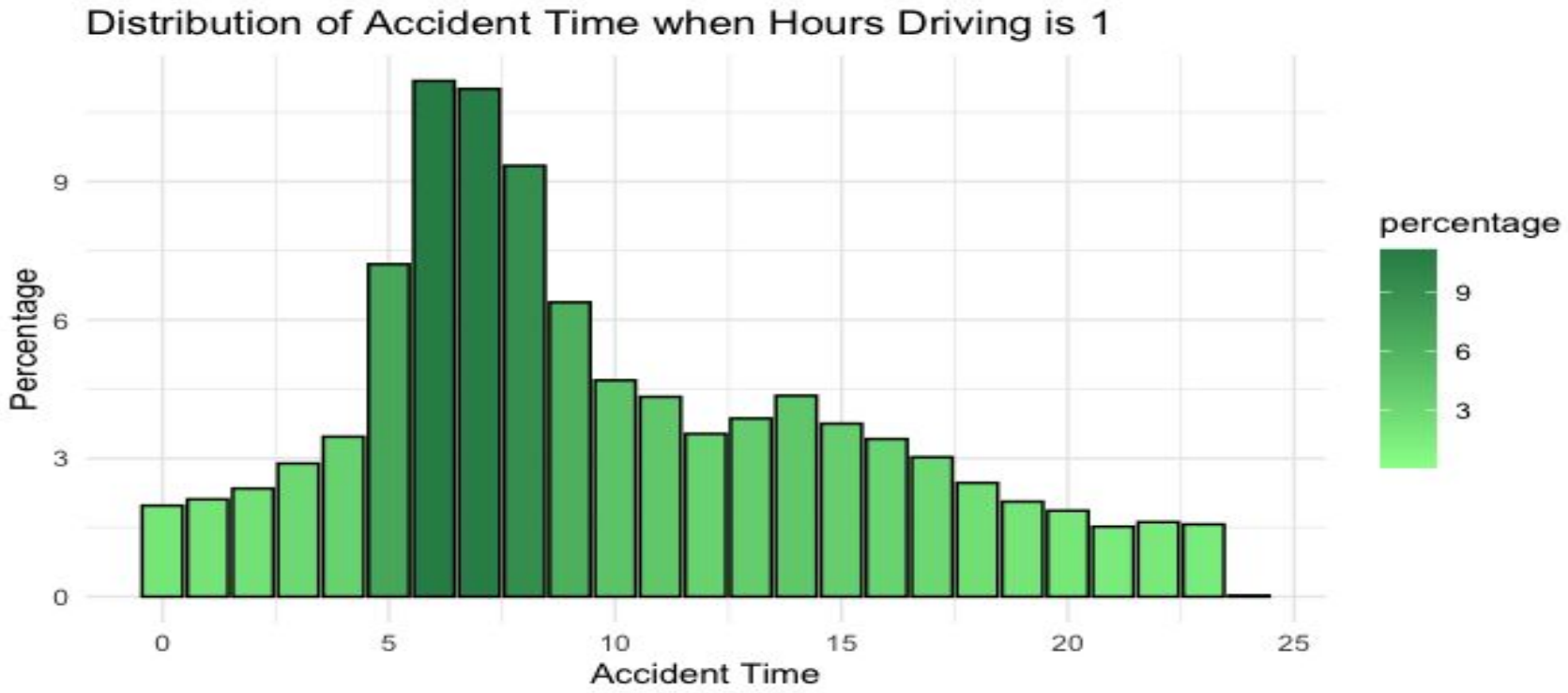

| | |
|---|---|
| 1 | Daylight |
| 2 | Dark |
| 3 | Dark, but lighted |
| 4 | Dawn |
| 5 | Dusk |

# Condition of the Atmosphere during Accidents

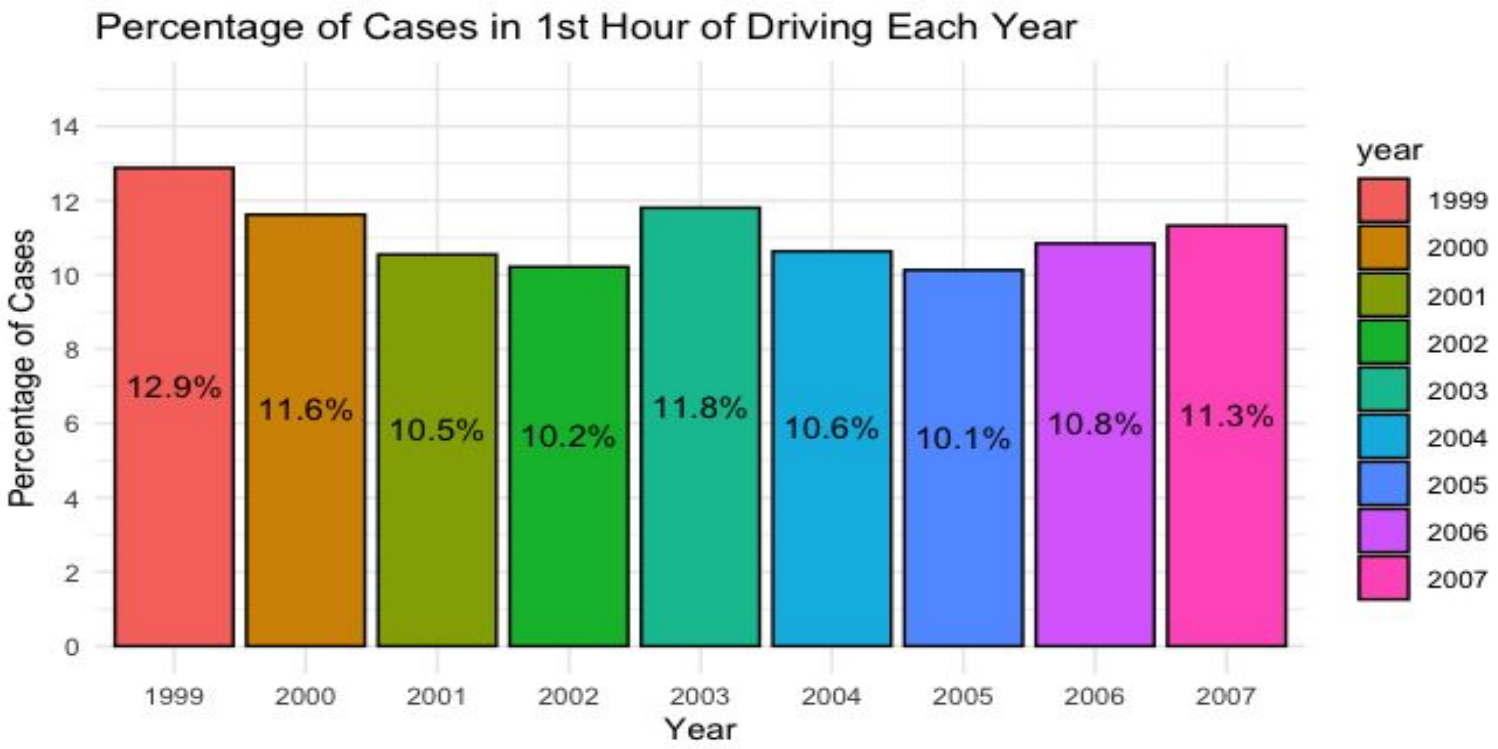



| | |
|---|---|
| 1 | Normal |
| 2 | Mist |
| 3 | Sleet |
| 4 | Snow |
| 5 | Fog |
| 6 | Rain + Fog |
| 7 | Sleet + Fog |
| 8 | Other |

# Occasions of Accidents



# Annual count during the 1st Driving Hour



# What do we get from the 1st Hour Data?

**Rate of Fatal Accidents:** About 30% of the accidents went down in the first hour when truck drivers kicked off their journeys.

**Light Condition:** Around 65% of accidents occurred during the daytime.

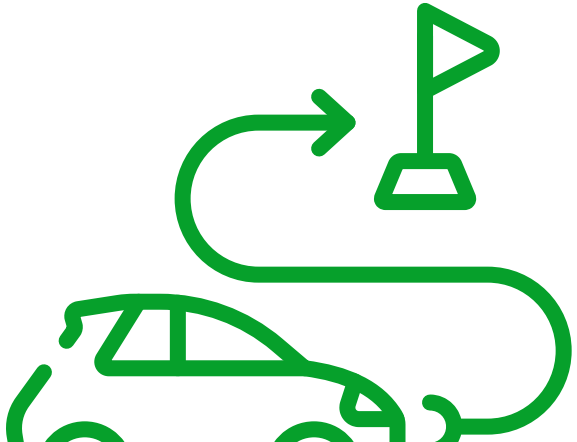
**Weather Condition:** The majority of accidents experienced no adverse weather conditions.

**Accident Timing:** Accidents mostly took place during the morning rush hour.

**Yearly Amount:** Cases experienced a minor fluctuation over the years.

# Contrasting Risks

The first hour of driving is the most dangerous, probably because accidents often happen when trucks are in the city or merging into traffic, and, these things tend to occur more in the first hour when truckers are just getting back on the road



# Conclusion

While the duration of continuous driving hours doesn't directly correlate with the rate of fatal accidents involving trucks, the Hours of Service Regulations still enforce driving time limits with breaks for truck drivers

# Thank You For Listening!

**Do you have any questions?**

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